

Projectile Motion: Prediction and Experimental Verification

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Abstract

This experiment investigated the projectile motion of a ball launched from a spring-loaded gun to predict its trajectory under gravitational acceleration. The initial velocity of the projectile was determined by measuring the horizontal range when launched from a known height and angle. Multiple trials at measured distances were used to determine initial velocity. Using kinematic equations, we found the initial velocity. This velocity was then used to predict the horizontal position for a given launch angle and height. The experimental method successfully demonstrated the independence of horizontal and vertical motion components, validating classical mechanics principles for two-dimensional motion under constant acceleration.

Scope

This report addresses the fundamental physics question: Can the trajectory of a projectile be accurately predicted using kinematic equations when only initial conditions are known? The analysis demonstrates practical application of two-dimensional motion equations for engineering students and physics practitioners who need to understand ballistic trajectories. The intended audience includes undergraduate physics students and instructors seeking to verify theoretical predictions through experimental measurement.

Recommendations

Implement angular measurement protocols. Future experiments should incorporate a digital inclinometer or protractor with finer angular resolution to reduce uncertainty in the launch angle measurement, as this parameter significantly affects vertical velocity component calculations.

Improved distance and height measurements. Future experiments using the spring-loaded gun would benefit from an integrated system of measurement for determining the height of the barrel. Additionally, an indicator for alignment of the instrument used to measure horizontal distance would improve precision.

Extend analysis to air resistance effects. Subsequent investigations should compare predicted trajectories with actual paths at higher velocities or

with projectiles of different cross-sectional areas to quantify when air resistance becomes non-negligible and requires drag force corrections to the idealized model.

Procedure, Materials, and Methods

The experiment utilized a spring-loaded projectile launcher from a Ballistic Pendulum Apparatus, a spherical projectile, catch box, carbon paper for impact marking, and meter sticks for distance measurement.

The experimental procedure consisted of two phases: initial velocity determination and trajectory prediction verification. In the first phase, the launcher was positioned horizontally at the edge of a laboratory table facing inwards with the muzzle height measured as 0.24 meters above the table to establish the initial vertical position. The projectile was launched horizontally into a catch box positioned on the table alongside the meter stick, with the landing position marked using carbon paper. This process was repeated for five trials to obtain consistent range measurements. The horizontal distance from the launch point to each impact position was measured and recorded (Table 1). The average horizontal range and standard deviation were calculated from these measurements to characterize the launch consistency and provide uncertainty estimates. Using the kinematic equation for vertical motion under constant acceleration, the time of flight was calculated from the vertical displacement and gravitational acceleration of 9.8 meters per second squared. The initial horizontal velocity was then determined by dividing the average horizontal range by the time of flight, with uncertainty propagated through the calculation.

In the second phase, the launcher was manipulated so as to create an angled trajectory. The launch angle was provided by the instructor as 30° and set using the integrated protractor. The initial height above the table surface was measured at the muzzle position. Using the previously determined initial speed and the measured launch angle, the horizontal and vertical velocity components were calculated using trigonometric relationships. With these values – initial velocities, gravity, and launch angle – kinematic equations were used to calculate the expected range with a quadratic.

Findings

The horizontal launch trials yielded range measurements that demonstrated reasonable consistency across repeated launches. Table 1 presents the complete dataset from the initial velocity determination phase.

Table 1: Horizontal Range Measurements for Initial Velocity Determination

Trial	Horizontal Range (m)
1	0.65
2	0.60
3	0.60
4	0.67
5	0.64
Mean	0.632
Std Dev	0.029

The average horizontal range was calculated as 0.632 meters with a standard deviation of 0.029 meters, corresponding to a coefficient of variation of 4.6 percent. This relatively small variation indicates good repeatability in the spring launcher mechanism and consistent projectile release conditions. The initial vertical height from the barrel to the table was measured at 0.24 meters. Using the kinematic equation for projectile motion, the initial velocity was calculated as

$$v_0 = x_f * \sqrt{(g / 2y_0)} = 0.632 \text{ m} * \sqrt{(10 \text{ m/s}^2 / 2 \times 0.24 \text{ m})} = 2.88 \text{ meters per second.}$$

For the angled launch configuration, the spring-loaded gun was adjusted to create an inclination angle of 30 degrees above the horizontal. This angle was specified by the instructor for the prediction challenge. The initial launch height above the table surface was measured at the muzzle position. The horizontal and vertical components of the initial velocity were calculated using $v_{x0} = v_0 \cos(\theta) = 2.88 \text{ m/s} \times \cos(30^\circ) = 2.48 \text{ m/s}$ and $v_{y0} = v_0 \sin(\theta) = 2.88 \text{ m/s} \times \sin(30^\circ) = 1.43 \text{ m/s}$, where $v_0 = 2.88 \text{ m/s}$ is the initial speed determined in the first phase and $\theta = 30^\circ$ is the launch angle. These velocity components, combined with the initial position coordinates, provided all parameters necessary for trajectory prediction.

The prediction task required determining the horizontal distance at which the projectile would strike the ground. Using the equation $y_f = -g/2v_0^2 * 1/\cos(\theta)^2 * x_f^2 + \tan(\theta) * x_f + y_0$, we find the value for x final.

where $y_f = 0$ (ground level), $g = 9.8 \text{ m/s}^2$, $v_0 = 2.88 \text{ m/s}$, $\theta = 30^\circ$, $y_0 = 0.27$.

Rearranging this equation into standard quadratic form:

$$(-g/2v_0^2\cos(\theta)^2) * x_f^2 + \tan(\theta) * x_f + (y_0 - y_f) = 0$$

Substituting the known values:

$$(-9.8/2(2.88)^2\cos(30^\circ)^2) \times x_f^2 + \tan(30^\circ) \times x_f + y_0 = 0$$

This yields:

$$-0.8 \times x_f^2 + 0.577 \times x_f + 0.27 = 0$$

Solving for x_f using the quadratic formula, and we predict that $x_f = 1.04$.

Conclusions

This experiment successfully demonstrated the predictive power of kinematic equations for projectile motion under ideal conditions. The two-phase experimental approach—first determining initial velocity through horizontal launch measurements, then applying this value to predict an angled trajectory—validated the fundamental principle that horizontal and vertical motion components remain independent under constant gravitational acceleration.

The initial velocity determination phase yielded consistent results across five trials, with a mean horizontal range of 0.632 meters and a coefficient of variation of only 4.6 percent. This repeatability confirms the reliability of the spring-loaded launcher as an experimental apparatus and demonstrates that systematic measurement techniques can produce reproducible data despite inherent experimental uncertainties. From these measurements, the initial projectile velocity was calculated as 2.88 meters per second, a value that served as the foundation for subsequent trajectory predictions.

The angled launch prediction, calculated using kinematic equations with the experimentally determined initial velocity, yielded an expected horizontal range of 1.04 meters for a 30-degree launch angle and initial height of 0.27 meters above the table surface. The ability to predict this landing position using only measured initial conditions and fundamental physics principles demonstrates the practical utility of two-dimensional motion analysis for engineering and scientific applications where trajectory prediction is essential.

The experimental methodology validates several key physics concepts. First, the independence of horizontal and vertical motion components allows separate treatment of each dimension, simplifying complex trajectory analysis into manageable calculations. Second, the constant acceleration model (gravitational acceleration of 9.8 m/s^2) provides accurate predictions at the velocity scales examined in this experiment, suggesting negligible air resistance effects for low-speed projectiles. Third, the successful application of the quadratic trajectory equation confirms that projectile paths follow parabolic trajectories as predicted by Newtonian mechanics.

The limitations identified in this experiment—primarily concerning angular measurement precision and height determination accuracy—represent opportunities for methodological refinement rather than fundamental flaws in the theoretical framework. The recommendations for improved instrumentation and extended analysis to include air resistance effects would enhance the quantitative precision of future investigations while maintaining the core experimental design that proved effective in this study.

Overall, this experiment fulfills its pedagogical objective of demonstrating how theoretical physics principles translate into practical predictive capabilities, providing undergraduate students with hands-on experience in experimental

design, data collection, uncertainty analysis, and the validation of mathematical models through empirical measurement.